$P(y=0) = 3/8$ — (0.5) $\quad P(y=1) = 5/8$ — (0.5)

$P(x_1=0|y=0) = \frac{1}{3} \rightarrow$ (1)

$P(x_2=0|y=0) = \frac{2}{3} \rightarrow$ (1)

$P(x_1=0|y=1) = \frac{3}{5} = 1 \rightarrow$ (1)

$P(x_2=0|y=1) = \frac{2}{5} \rightarrow$ (1)

$$P(y=0|x_1=0, x_2=0) = \frac{P(x_1=0, x_2=0|y=0)P(y=0)}{P(x_1=0, x_2=0)}$$

$$= \frac{P(x_1=0|y=0)P(x_2=0|y=0)P(y=0)}{P(x_1=0)P(x_2=0)}$$

$$= \frac{1/3 \times 3/8 \times \frac{2}{3}}{\frac{1}{3} \times \frac{3}{8} \times \frac{2}{3} + 5/8 \times 1 \times 2/5}$$

$$= \frac{\frac{1}{12}}{\frac{1}{12} + 1/4} = 1/4 \rightarrow (1.5)$$

$$P(y=1|x_1=0, x_2=0) = \frac{5/8 \times 1 \times 2/5}{1/3 \times \frac{3}{8} \times \frac{2}{3} + 5/8 \times 1 \times 2/5}$$

$$= \frac{1/4}{1/12 + 1/4} = 3/4 \rightarrow (1.5)$$

$$P(y=0 \mid x_1=0, x_2=1) = \frac{P(x_1=0 \mid y=0)\, P(x_2=1 \mid y=0)\, P(y=0)}{P(x_1=0, x_2=1)}$$

$$= \frac{3/8 \times \frac{1}{3} \times 1/3}{3/8 \times \frac{1}{3} \times \frac{1}{3} + 5/8 \times 1 \times 3/5} = \frac{1/24}{\frac{1}{24} + 3/8}$$

$$= \frac{1}{10} = 0.1$$

$$P(y=1 \mid x_1=0, x_2=1) = 1 - P(y=0 \mid x_1=0, x_2=1)$$

$$= 1 - 0.1 = 0.9 \rightarrow (2)$$

KNN Regression.

1)

x	Actual y	Predicted y
1	1	1
4	2	2
5	5	5

$$MSE = \frac{1}{3}\left((1-1)^2 + (2-2)^2 + (5-5)^2\right)$$

$$= \frac{1}{3}(0) = 0 \quad \text{——} (2)$$

2)

x	Actual y	Predicted y
1	1	2
4	2	5
5	5	2

$$MSE = \frac{1}{3}\left[(1-2)^2 + (2-5)^2 + (5-2)^2\right]$$

$$= \frac{1}{3}\left[1 + 9 + 9\right] = 6.33 \longrightarrow (3)$$

Question 3 :

$$H(Y) = -\frac{5}{8}\log_2 5/8 - 3/8 \log_2 3/8$$

$$= 0.9544 \longrightarrow (1)$$

$$P(X_1 = T) = \frac{4}{8} \qquad P(X_1 = F) = \frac{4}{8}$$

$$H(Class = T \mid X_1 = T) = -\frac{4}{4}\log\frac{4}{4} - \frac{0}{4}\log 0/4 = 0 \quad (0.5)$$

$$H(Class = T / X_1 = F) = -1/4 \log 1/4 - 3/4 \log 3/4 = 0.8112$$

$$H(Class / X_1) = \frac{1}{2} \times 0 + \frac{1}{2} \times 0.8112 \quad (0.5)$$

$$= 0 + 0.8112 \longrightarrow (0.5)$$

$$IG = 0.9544 - 0.8112 \longrightarrow (0.5)$$

$$= 0.5488$$

$$P(X_2 = T) = \frac{4}{8}$$

$$P(X_2 = f) = 4/8$$

$$H(Class \mid X_2 = T) = -3/4 \log 3/4 - \frac{1}{4}\log 1/4$$

$$= 0.8113 \rightarrow (1)$$

$$H(Class / X_2 = f) = -2/4 \log 2/4 - 2/4 \log 2/4$$

$$= 1 \quad (1)$$

$H(Class/X_2) = \frac{4}{8}(0.8113) + \frac{4}{8}(1)$

$= 0.9056 \longrightarrow (0.5)$

$IG = 0.9544 - 0.9056 = 0.0488 \longrightarrow (0.5)$

c) Since IG of X_1 > IG of X_2 hence it will provide the best split. $\longrightarrow (1)$